

“What is the value of {templates}?” Rethinking Document Information Extraction Datasets for LLMs

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Abstract

The rise of large language models (LLMs) for visually rich document understanding (VRDU) has kindled a need for prompt-response, document-based datasets. As annotating new datasets from scratch is a labor-intensive task, the existing literature has generated prompt-response datasets from available resources using simple templates. For the case of key information extraction (KIE), one of the most common VRDU tasks, past work has typically employed the template “What is the value for the {key}?”. However, using simple and uniform templates is insufficient for creating robust models in both research and industrial contexts, given the variety of questions encountered in the wild. In this work, we present K2Q, a publicly available collection of five datasets that have been converted from KIE to a prompt-response format using a plethora of bespoke templates. The questions in K2Q can span multiple entities and be extractive or boolean. We empirically compare the performance of seven baseline generative models on K2Q with zero-shot prompting. We further compare three of these models when training on K2Q versus training on simpler templates to motivate the need of our work. We find that creating diverse and intricate KIE questions enhances the performance and robustness of VRDU models, and hope that this work encourages future study on data quality for generative model training.

1 Introduction

Visually rich document understanding is a core research field at the intersection of natural language processing (NLP) and computer vision. The field aims to develop methods to process information from documents and perform natural language inference on it. Naturally, diverse and high-quality datasets are crucial for training and evaluating VRDU models. This is ever more important as generative models, such as large language models, are becoming the mainstream state-of-the-art

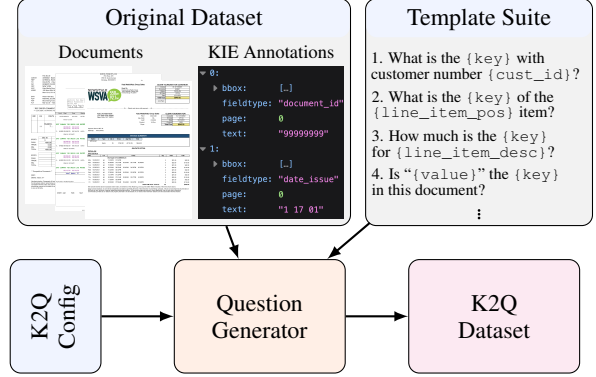


Figure 1: Generation pipeline of K2Q datasets. A suite of diverse templates is designed for each specific KIE dataset. These templates are populated in accordance to a configuration file that configures the dataset size and proportion of extractive and assertive questions.

method for tackling VRDU problems (Bai et al., 2023; Wang et al., 2023a; Hu et al., 2024). These models require significant amounts of data for training as well as a diverse test set to evaluate whether robust document understanding is achieved.

VRDU covers an array of tasks which can be used for training generative models. While visual question answering (VQA) datasets (Mathew et al., 2021; Landeghem et al., 2023) can be directly used for instruction tuning, only a handful of Document VQA datasets are publicly available (Mathew et al., 2021; Landeghem et al., 2023). Thus, existing datasets for other tasks are generally transformed into a prompt-response style to increase training volume and document diversity.

Current works that leverage existing non-VQA VRDU datasets for generative models often populate uniform templates (Wang et al., 2023a; Ye et al., 2023b; Tanaka et al., 2024), resulting in datasets that lack diversity and complexity. Specifically, for key information extraction (KIE), which aims to find important entities within a document, datasets tend to populate the template, “What is the value for the {key}?” (Ye et al., 2023a,b; Hu et al.,

2024).¹ While such questions directly translate the KIE problem, they fail to capture the complexities of VRDU and the intricacies of real-world applications of generative models for documents.

In this paper, we propose a new collection of transformed KIE datasets, **K2Q**, that aims to find a balance between templated datasets and human-annotated VQA datasets. K2Q is derived from five datasets: CORD (Park et al., 2019), Docile (Simsa et al., 2023), Kleister Charity (Stanislawek et al., 2021), and VRDU Ad-Buy (Ad-Buy) and Registration Form (Reg. Form) (Wang et al., 2023c). We curate over 100 different templates that lead to a diverse set of over 300,000 questions across over 12,000 documents. The transformed datasets contain extractive questions as well as boolean (i.e., true or false) questions, where questions may consider multiple entities within a document. Our key contributions are given below:

- We introduce a new publicly available collection of datasets, K2Q, that converts five existing KIE datasets into rich and diverse prompt-response datasets.² The creation pipeline for K2Q is illustrated in Figure 1.
- We show that K2Q exhibits closer characteristics to human-made VQA datasets than simple templates through substantially lower self-Bleu and perplexity scores.³
- We provide zero-shot and fine-tuned benchmarks for K2Q across seven models.⁴
- We conduct a detailed analysis of the impact of diverse, dataset-specific templates on VRDU model performance and groundedness against simple templates. We find that training on diverse templates improves relative ANLS performance by 40% over simple templates.

2 Key Information Extraction in VRDU

We focus on KIE in this work due to the abundance of existing datasets in the VRDU literature and the importance of information extraction in industry.⁵ KIE is the task of identifying important entities

¹For example, finding an address on a form is converted into the question “What is the value for the address?”

²A subset of K2Q is provided in the supplementary materials. We will release the full dataset publicly upon publication.

³Perplexity of observing DocVQA and DUDE questions.

⁴Additionally, we provide a detailed performance breakdown in App. D.2.

⁵We hope future work will examine other popular structured VRDU tasks such as document classification (Harley et al., 2015) or document structure prediction (Li et al., 2020; Xu et al., 2022; Zmigrod et al., 2024a).

Dataset	Docs	Pages	Labels	Lines
Ad-Buy	641	1,598	14	✓
CORD	1,000	1,000	33	✓
Docile	5,680	7,372	55	✓
KLC	2,778	62,010	8	✗
Reg. Form	1,915	3,890	6	✗
BuDDIE	1,665	1,665	69	✗
DeepForm	1,100	4,720	5	✗
FUNSD	199	199	4	✗
KNDA	540	3,229	4	✗
SROIE	600	1,000	4	✗
WildReceipt	1,786	1,786	25	✗

Table 1: KIE Datasets. The first section contains the datasets we transform in K2Q. The second section contains other relevant KIE datasets. The *Lines* column indicates whether the dataset contains line items.

within a document. As such, many datasets have naturally examined invoices and receipts (Huang et al., 2019; Park et al., 2019; Sun et al., 2021; Simsa et al., 2023), where entities are clearly defined and often connect through **line items** (e.g., the name and price of an item appear on the same “line”). Similarly, forms and legal documents have also been the focus of a plethora of datasets (Jaume et al., 2019; Borchmann et al., 2021; Stanislawek et al., 2021; Wang et al., 2023c; Zmigrod et al., 2024b). We provide details regarding available KIE datasets in Table 1.⁶

2.1 KIE for Generative Models

In order to train generative models for KIE, datasets must be formulated into a prompt-response type structure. So far, the VRDU and NLP literature has proposed three main strategies to obtain such formulations: populating simple templates based on existing annotations, manually curating questions, or generating questions using LLMs. We discuss the advantages and disadvantages of each method below to motivate our approach in Section 3.

Populating Simple Templates. The most common method for designing large datasets of instructions for LLMs is templating. This is the case in both general NLP (Chung et al., 2022; Wei et al., 2022) as well as in VRDU (Tang et al., 2023; Ye et al., 2023b; Zhang et al., 2023; Wang et al., 2023c; Tanaka et al., 2024; Zmigrod et al., 2024b). Unfortunately, most templating approaches in VRDU rely on simplistic and dataset-agnostic templates that do not reflect a model’s true understanding of

⁶We only focus on English datasets in this work. Nevertheless, other non-English VRDU datasets exist (Wang et al., 2021; Qi et al., 2022; Ding et al., 2023).

Ad-Buy	Q: When does the advertisement start running on channel KPTH on program "People's Court"? A: 01/30/20 Q: Did the advertisement cost Teresa Tomlinson for Senate \$1,855.00? A: Yes
CORD	Q: How much did the order(s) of Combo 1 cost in total? A: 30.000 Q: Is "20.000" the number of servings/quantity of a menu item in this receipt? A: No
Docile	Q: How much is the total amount with tax of the 19th item? A: 130.00 Q: Is "Alyse For Alaska" the name of the customer that is being invoiced in this document? A: No
KLC	Q: How much did the charity with number 1154288 earn annually in British pounds? A: 36493079.00 Q: Is 287408.00 pounds the annual expenditure of Campaign For Learning? A: Yes
Reg. Form	Q: Who is the agent of the form? A: Hogan & Hartson LLP Q: Is manager the title of James A. Coppola? A: No

Figure 2: Examples of populated questions and answers from K2Q.

the task nor the real-life queries that would be submitted to a generative model. For instance, Ye et al. (2023a,b); Wang et al. (2023a); Hu et al. (2024) invoke a single simple template: "What is the value for the {key}?". This can lead to under-specified questions, which our work aims to avoid. BuDDIE (Zmigrod et al., 2024b) is similar to these works but also includes boolean questions. In comparison, our work populates much richer templates thanks to the inclusion of multiple rephrasings per question, multiple entities per question, and multiple question types. Another approach to templating is that of the InstructDoc (ID) collection (Tanaka et al., 2024), which features five rephrased templates per dataset that ask the model to classify text snippets into one of the key entity types. Unfortunately, we believe this to be unrepresentative of the KIE task in practice, as the underlying goal of KIE is to *directly extract* entities from a large amount of text, instead of simply classifying candidate extractions. Therefore, we do not include such templates here.

Manually Constructing Questions. Rather than relying on existing datasets from other tasks, researchers have also constructed prompt-response datasets from the ground up, such as DocVQA (Mathew et al., 2021; Tito et al., 2022) and DUDE (Landeghem et al., 2023). Due to the lack of other VQA datasets truly focused on visually rich documents, the literature has also considered datasets from adjacent areas such as InfographicsVQA (Mathew et al., 2022) or ChartQA (Masry et al., 2022). The benefits of human-annotated datasets come from the quality, diversity, and depth of the questions that can be achieved. Unfortunately, the cost of curating VQA datasets is high, and thus creating datasets of the scale needed to instruction-

tune LLMs can be inaccessible to many practitioners. Our work addresses this by applying human intervention at the dataset level rather than at the document level. We use annotators to manually design templates, resulting in a collection of datasets over 6 and 7 times larger than DocVQA and DUDE respectively.⁷

Generating Questions using LLMs. Past work has demonstrated that utilizing LLMs to create fine-tuning datasets may be a promising research avenue (Honovich et al., 2023; Peng et al., 2023; Taori et al., 2023; Wang et al., 2023b). This also includes work for general multi-modal LLMs (Liu et al., 2023; Li et al., 2023), suggesting this methodology can extend to VRDU. However, to the best of our knowledge, the VRDU literature has not investigated the use of LLMs to generate prompt-response pairs yet. While we do not leverage LLMs to generate data in this work, we believe a future iteration of K2Q could use an LLM to augment our set of templates.

2.2 Modeling Spectrum for KIE

Models for VRDU (and thereby KIE) lie on a spectrum of those that only utilize text extracted from documents through optical character recognition (OCR) (Devlin et al., 2019; Liu et al., 2019) to those that ingest documents directly as image inputs (Lee et al., 2023). Naturally, works also lay in the middle, with some more focused on OCR and spatial features (Xu et al., 2020; Peng et al., 2022) and some others more focused on complementing a main visual channel with text (Kim et al., 2022;

⁷This size factor is after we sample our selection of templates. One can increase the sampling ratio to make a significantly larger dataset.

Dataset Name	Num. Temp.	Num. Ques.	% Extr. Ques.	% 1-Page Ques.	Ques. Length	Answer Length	Ent. per Ques.	Ques. per Doc.
Ad-Buy	50	15,119	57.8	96.8	9.5	2.3	1.5	23.6
CORD	22	39,575	49.6	100.0	12.6	1.4	1.4	39.6
Docile	17	185,557	68.1	99.9	11.6	3.6	1.1	32.7
KLC	31	44,813	43.2	88.3	9.8	1.6	1.8	16.1
Reg. Form	18	23,427	37.7	99.9	8.8	3.1	1.1	12.2

Table 2: K2Q statistics per dataset. Question length and answer length show the average number of tokens of questions and answers respectively.

Davis et al., 2022; Tang et al., 2023). Multi-modal LLMs have also been developed along this spectrum (Wang et al., 2023a; Ye et al., 2023b,a; Hu et al., 2024; Tanaka et al., 2024).

3 The K2Q Dataset

We present **Key Information Extraction Transformed to Visual Question Answering Datasets**, or **K2Q** for short. This is a collection of five KIE datasets transformed into prompt-response type datasets (similar to VQA), namely Ad-Buy, CORD, Docile, Kleister Charity, and Reg. Form. These datasets were chosen as they cover a wide range of domains within the field of VRDU. K2Q is publicly available⁸ and contains more than 300,000 questions over 12,000 documents. The questions are designed to reflect the intricacies of document understanding such as co-references, disambiguation of similar entities, and questions spanning multiple pages. Figure 2 provides several examples from each dataset of K2Q. Table 2 shows a statistical breakdown of the questions generated for K2Q.⁹

3.1 Dataset Construction

Unlike other template-based approaches, we manually construct several templates *at the dataset level* for each entity in the KIE topology.¹⁰ When possible, questions are created in relation to other entities as shown in Figure 2 (e.g., “Is manager the title of James A. Coppola?” involves two distinct key entities from Reg. Form).¹¹ In particular, CORD, Docile, and Ad-Buy contain line items that connect several different entities in an entry (similar to the rows of a table). Examples of such items are lines in a receipt and advertisement slots. Furthermore, K2Q contains **extractive** and **boolean**

questions; extractive questions have answers that are key entity values, while boolean questions have answers that are “Yes” or “No”. To ensure the quality of the questions generated, we address several complexities.¹²

Cleaning OCR Output. In the traditional KIE task, predictions are typically given as OCR token spans which can be noisy. It is thus necessary to clean the OCR entity values to enable generative models trained on K2Q to produce more natural responses. We provide more details regarding our data cleaning in App. A.1. Note that the OCR output provided in the original datasets is used to train and test the OCR-based models in Section 4.

Under-specified Questions. It is sometimes possible for a question to have several correct answers within a document. K2Q handles such cases in two different ways. Firstly, if a document contains multiple entities of the same type that are not line items (e.g., several vendor names mentioned), we consider all entities to be allowed as an answer.¹³ Secondly, if multiple line item rows contain repeated information, we avoid asking questions that may cause ambiguity when trying to determine the row to consider. For example, if a receipt contains two items that both cost \$10, we do not include the question “Which item in the receipt cost \$10?” as it is unclear which line item is being referred to.¹⁴

Negative Boolean Question Generation. Half of all boolean questions are designed to be false. To do so, we devise a candidate set of false answers by drawing inspiration from the approach introduced in Zmigrod et al. (2024b). Firstly, we

⁸Use of K2Q follows the same terms as the datasets it transforms. For more details, see the license of each dataset.

⁹We also provide the data split sizes in App. A.2.

¹⁰Templates were created by three VRDU researchers who were already familiar with the datasets. The datasets were split evenly among them for template creation and validation.

¹¹K2Q ensures that all entities in the question and answer are present on the page given to models.

¹²While K2Q is intended for training generative models, we ensure it can be used as a VQA training dataset for traditional VRDU models such as (Xu et al., 2020). This required some heuristics to be used for Docile and KLC, which do not relate KIE entities to the OCR.

¹³This under specificity only occurs for Docile.

¹⁴While it may be desirable for models to learn to return all relevant line items in this case (multi-span answer), it complicates training evaluation so we omit these questions. Future work will look into incorporating such questions.

Property	DUDE	ID	UReader	K2Q
Templated	✗	✓	✓	✓
Diverse	✓	✗	✗	✓
Extractive	✓	✗	✓	✓
Boolean	✓	✗	✗	✓
Unambiguous	✓	✓	✗	✓

Table 3: Properties of VRDU datasets for generative models. DUDE is provided here as a human-generated reference.

Error Type	Error rate	Fleiss κ
Template	1.15%	0.53
Cleaning	1.38%	0.31
Annotation	2.76%	0.51
Other	0.23%	0.51
Total	5.52%	0.49

Table 4: K2Q validation results. We breakdown errors into four categories: (1) Template errors indicate a question design issue (2) Cleaning errors indicate a data cleaning issue (3) Annotation errors indicate issues with the original datasets (4) Other errors (e.g., OCR issues).

consider all unique values of a given entity type in the dataset. Secondly, for datasets that have a hierarchical entity ontology, we also consider unique values in the same document that share a parent entity. Lastly, we additionally consider other values in the document that have the same format (e.g., string, integer, date, etc.). The false candidate is then sampled from one of these three sets.

Question Sampling. Due to the nature of templates, we generate a finite number of questions for each document to avoid introducing too much redundancy. For Ad-Buy, Kleister Charity, and Reg. Form, this was achieved by randomly sampling one extractive question per entity and then a fixed number of boolean questions. Due to the high number of entities for CORD and Docile, we generate all questions and sample after generation to select a fixed number of questions in both the extractive and boolean settings.

Dataset Validation. K2Q is generated systematically after manually curating templates; nevertheless, we applied various forms of validation to ensure a high-quality dataset. Prior to question generation, every template was reviewed by at least one other template writer to verify grammatical correctness and variety. Post question generation, five documents from each dataset were randomly sampled. For each document, up to ten extractive and ten boolean questions are sampled, and three validators check questions for grammatical correctness and other data issues. An issue was considered if

Source Dataset	New Dataset	Num. Temp.	Num. Ques.	Ques. per Doc.
Ad-Buy	K2Q	50	15,119	23.6
	SD	1	4,986	7.8
CORD	K2Q	22	39,575	39.6
	SD	1	4,143	4.1
	ID	5	1,336	1.3
Docile	K2Q	17	185,557	32.7
	SD	1	53,547	9.4
	ID	5	56,369	9.9
KLC	K2Q	31	44,813	16.1
	SD	1	19,348	7.0
	ID	5	13,449	4.8
	UReader	1	22,224	8.0
Reg. Form	K2Q	18	23,427	12.2
	SD	1	8,826	4.6

Table 5: Comparison of variety of questions in K2Q compared to SD, ID, and UReader.¹⁵ We only give statistics for datasets that are featured in this work, though ID and UReader also use other datasets. Examples for each dataset are given in App. B

two out of three validators noted it. The validation results are given in Table 4. The overall Fleiss-kappa scores indicate moderate agreement between raters. The disagreements demonstrate the complexity of assessing erroneous questions. We also note that half of the errors are due to annotation issues with the original datasets, and so inevitably occur in K2Q.

3.2 Baseline datasets

Admittedly, K2Q requires more care and effort to collate than past template-based datasets. We empirically motivate why this additional work is worthwhile by comparing results against a simple template approach. To do this, for each of the five datasets in K2Q, we construct a simple dataset where we use the template “What is the value for the {key}?” for all entities that are not line items¹⁶. We refer to this **simple** collection of datasets as **SD**.

3.2.1 Comparison to Related Datasets

Table 3 summarizes the properties of different datasets. A comparison between the questions generated for each dataset in K2Q is provided in Ta-

¹⁵Exact details of the dataset size for UReader are not shared. The numbers in the table were inferred from the description in Ye et al. (2023b).

¹⁶We do not create line item questions with this template as they would be ambiguous for models. For example, if a receipt has multiple items, the question “What is the value for the receipt item?” is too ambiguous to answer. This can be fixed by replacing “the” with “any”.

Dataset	Perplexity ($\downarrow$)		Self-BLEU ($\downarrow$)	
	DUDE	DocVQA	2-gram	4-gram
DUDE	28.5	35.9	0.73	0.40
DocVQA	45.1	27.4	0.83	0.58
K2Q	229.6	228.6	0.92	0.83
SD	1592.8	928.5	1.00	1.00

Table 6: Perplexity and Self-BLEU scores for DUDE, DocVQA, K2Q, and SD.

ble 5. We give examples of questions generated in ID and UReader in App. B.

To quantitatively assess how closely K2Q resembles human-annotated datasets in comparison to SD, we conduct two studies. First, we measure the perplexity of the questions of the DocVQA and DUDE test sets with a small language model (GPT2) fine-tuned on the question sets of K2Q and SD. Perplexity indicates how likely a language model is to generate new input and so a low perplexity indicates the input aligns with the model’s distribution. Secondly, following Ye et al. (2022), we compute the self-BLEU score of the questions of the K2Q and SD test sets to compare their diversity to that of DUDE and DocVQA questions.¹⁷ Table 6 gives the perplexity scores of GPT2 fine-tuned and tested on each combination of datasets as well as self-BLEU scores.¹⁸ We observe that, even though DUDE and DocVQA generalize well to each other, K2Q and SD have a much higher perplexity. Similarly, the self-BLEU scores of K2Q and SD are higher than those of DUDE and DocVQA – highlighting the challenges of mimicking human-crafted datasets using templates in general. However, fine-tuning on K2Q questions did reduce the perplexity from fine-tuning on SD by a factor of 4 and 7 on DocVQA and DUDE respectively. Additionally, the self-BLEU score of K2Q is substantially closer to that of DocVQA and DUDE than SD. These results demonstrate the benefits of K2Q over simpler templating methods in terms of similarity to human-curated data.

4 Experiments

4.1 Benchmark Models

To analyze the impact of training and testing on K2Q, we consider three non-LLM generative mod-

¹⁷See App. C.2 for more details on perplexity and Self-BLEU.

¹⁸For the perplexity experiments using K2Q and SD, we train the model for one epoch on all questions. To match the number of steps we train on, we fine-tune the model for three epochs on DocVQA and two epochs on DUDE.

Model	Ad-Buy	CORD	Docile	KLC	Reg-Form
Dnt _{zs}	1.4	18.0	7.2	3.8	5.9
P2S _{zs}	13.5	27.0	24.1	11.8	24.0
Doc _{zs}	23.9	43.0	48.0	80.4	27.6
UDOP _{zs}	29.1	29.7	35.1	30.0	39.1
mPD _{zs}	60.9	64.5	56.0	66.8	65.2
mPDC _{zs}	61.9	64.4	53.4	66.0	66.8
GPT4 _{zs}	72.7	85.3	61.3	68.1	76.5
Dnt _{sd}	7.7	24.4	22.3	15.6	19.7
P2S _{sd}	18.0	31.0	35.5	28.0	29.7
Doc _{sd}	46.3	43.6	53.0	62.8	66.2
Dnt _{mK2Q}	56.4	82.7	47.7	67.2	81.4
P2S _{mK2Q}	69.2	84.7	59.0	77.0	87.0
Doc _{mK2Q}	92.6	94.0	88.9	92.6	78.0
Dnt _{K2Q}	58.5	83.3	47.8	68.1	82.7
P2S _{K2Q}	73.8	86.5	59.5	79.4	88.7
Doc _{K2Q}	93.9	96.5	90.0	93.6	80.3

Table 7: ANLS ($\uparrow$) results on K2Q test set using various training settings for models Donut (Dnt), Pix2Struct (P2S), DocLLM (Doc), UDOP, mPlugDocOwl 1.5 (mPD), mPlugDocOwl 1.5-Chat (mPDC), and GPT4.

els in our experiments: Donut (200M parameters) (Kim et al., 2022) Pix2Struct base (282M parameters) (Lee et al., 2023), and UDOP (800M parameters) (Tang et al., 2023), which uses OCR-generated text input as well as the image of the document. Additionally, we consider four LLMs. Firstly, mPlugDocOwl 1.5 (8.1B parameters) and mPlugDocOwl 1.5-chat (8.1B parameters) (Hu et al., 2024), which are OCR-free LLMs and use the image of the document as input. Secondly, DocLLM (1.5B parameters) (Wang et al., 2023a) and the text-only variant of GPT4¹⁹ (OpenAI, 2023), which are OCR-based. GPT4, UDOP, and the mPlugDocOwl models are only run in the zero-shot experiments due to resource constraints and data contamination considerations. Indeed, UDOP and the mPlugDocOwl models incorporated some of our original datasets in their pretraining; this may also be the case for GPT4. We initialize trainable models from DocVQA fine-tuned checkpoints. Further model set-up details are given in App. C.1.

4.2 Evaluation Settings

We consider several training data settings for our experiments. Firstly, we focus on the full K2Q dataset as presented in Section 3. Secondly, a baseline dataset, SD (as described in Section 3.2), is used to quantify the importance of having richer and more diverse templates. K2Q offers two main

¹⁹GPT4 is prompted with text tokens only. All other models are prompted as described in their respective papers.

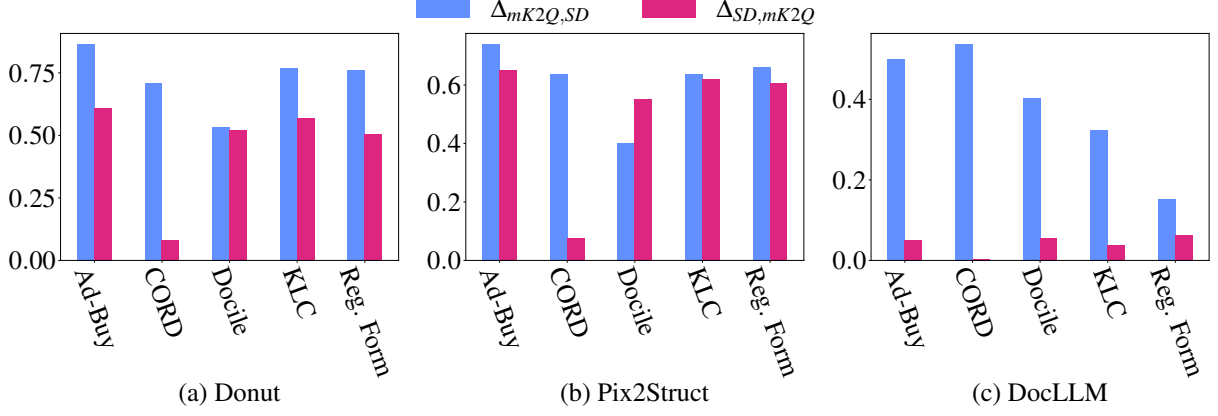


Figure 3: Comparison of training and evaluating on complex questions (mK2Q) and simple questions (SD).

advantages over SD: (1) the complexity of the questions and (2) the size of the dataset. To measure the impact of each advantage, we also train the models with a down-sampled version of K2Q which we call **mini-K2Q (mK2Q)**. This dataset aims to reflect the same complexities as K2Q but be comparable in size to SD. Note that questions that span multiple pages are not used in our experiments, as not all models have multi-page capabilities.

We use the Average Normalized Levenshtein Similarity (ANLS) metric (Biten et al., 2019) to evaluate model performance.²⁰ We choose ANLS as the performance measure because it focuses on the surface similarity of the generated and true answer (as is intended in extraction), but does not suffer from the rigidity of exact match.

4.3 Results on K2Q

Table 7 gives the benchmark results for K2Q.²¹ As expected, training on simple templates (SD) followed by testing on K2Q questions yields much lower performance than training directly on K2Q or mK2Q. We consider the effects of training and testing on different datasets in Section 4.4. In general, models trained on the simple templates perform slightly better than their zero-shot counterparts, except for DocLLM trained on KLC with simple templates and tested on K2Q templates. We also observe a consistent positive impact of using K2Q versus its down-sampled counterpart mK2Q across the board, highlighting the diversity of questions contained in K2Q as well as the impact of greater training volume enabled by a template-based approach. These trends support our hypothesized benefits of K2Q. App. D.2 contains an ablation study

on how the different training configurations impact performance on extractive and boolean questions, as well as on the number of entities present in each question.

We typically observe performance on the same data setting increases with model size. DocLLM is thus the best trainable model, followed by Pix2Struct and then Donut. In the zero-shot setting, GPT4 is the best model on all datasets but KLC. There is a large gap between the zero-shot and the fine-tuned results for Pix2Struct, Donut, and DocLLM despite the original checkpoints being fine-tuned on DocVQA. This indicates sensitivity to distribution shifts in the type of documents and questions.

4.4 Quantitative Comparison of Templates

This work hypothesizes that using the same simple template for all datasets does not reflect the intricacies of document understanding and may lead to reduced robustness. Instead, we propose a suite of rich and diverse templates for transforming each dataset to create K2Q. To validate this hypothesis and motivate the need for our work, we compare how models trained on simple templates perform when tested on K2Q templates, and **vice versa**. To evaluate this experiment, we define the following metric to quantify model robustness to change of question templates between train and test:

$$\Delta_{D_1, D_2} \stackrel{\text{def}}{=} \frac{\text{ANLS}_{D_1/D_1} - \text{ANLS}_{D_2/D_1}}{\text{ANLS}_{D_1/D_1}} \quad (1)$$

where D_1 and D_2 are datasets, and $\text{ANLS}_{D/D'}$ is the ANLS score of a model trained on the train split of dataset D and evaluated on the test split of dataset D' . In other words, Δ_{D_1, D_2} is the relative difference in performance for a model tested on D_1 after being trained on D_1 or D_2 .

²⁰We provide a definition of the ANLS score in App. C.2.

²¹Results on SD are given in App. D.1 for comparison.

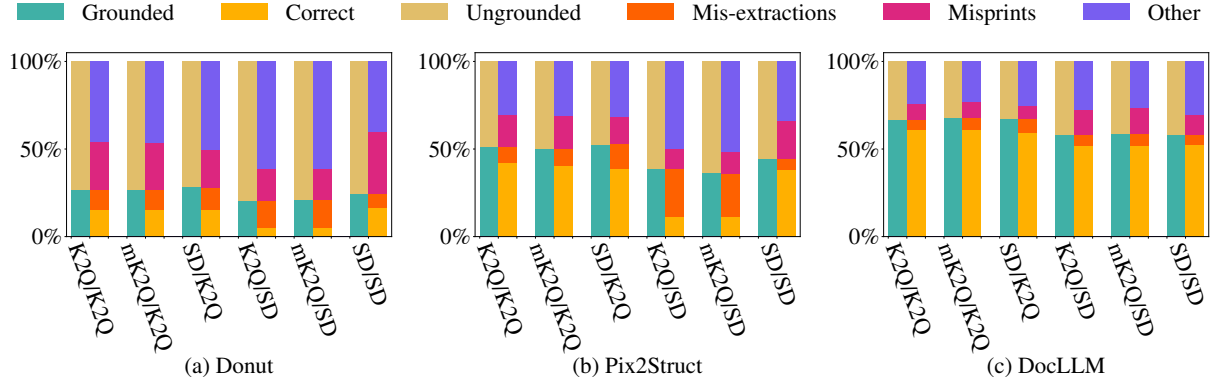


Figure 4: Detailed breakdown of groundedness and error types for KLC using different training / testing datasets.

Figure 3 shows the difference between $\Delta_{mK2Q,SD}$ and $\Delta_{SD,mK2Q}$ for the three trainable models. We use the down-sampled version of K2Q to enable a fair comparison based on template style not affected by training data volume. While training on the original dataset always outperforms training on the alternative dataset (seen by positive Δ values) due to input distribution shift, we note that training on our templates tends to yield better generalization to simple templates than the other way around. Indeed, for DocLLM, the average performance gain using K2Q over SD is 40% across the datasets. This observation corroborates the motivation behind K2Q that employing a rich and diverse set of templates for training models results in better robustness to new types of questions and formulations. We provide more metrics, including a comparison with the full K2Q in App. D.3.

4.5 Evaluation of Generated Errors

While ANLS is a useful metric for comparing extractive results from generative models, it fails to determine the cause of the errors, such as misreading the text in the document or not understanding the question. To determine this breakdown, we use the notion of **groundedness**. A generated response is considered grounded if it can be identified in the OCR output of the document.²² Correct generations are naturally grounded. We define incorrect generations that are grounded as **mis-extractions**. For ungrounded generations, we consider extracted strings that have an ANLS score of 0.8 or higher to be **misprints**. Any other errors we label as **other**. We provide examples in App. D.4.1.

Figure 4 gives the breakdown of groundedness

²²We consider a response to be in the OCR if we can find a non-case-sensitive match. The match can be partial.

and error types for the KLC extractive questions of K2Q using Donut, Pix2Struct, and DocLLM fine-tuned on K2Q. We choose KLC due to its large test set. The models tested on K2Q typically exhibit a higher level of grounding than those tested on SD, regardless of the training data. This could suggest that formulating templates for the dataset rather than using generic templates allows the models to contextualize the questions better. Grounded responses enable easier verification of KIE model outputs. Note that both Donut and Pix2Struct contain a large number of misprint errors compared to DocLLM. This could mean that using a more powerful OCR-free method may result in much stronger performance.

5 Conclusion

This paper introduced K2Q, a new publicly available collection of five transformed KIE datasets for generative VRDU models. K2Q provides a large and diverse set of questions that better capture the intricacies of KIE and the variety of questions that users can ask in real-world applications. Our experiments demonstrate that training generative models on K2Q instead of data from simple templates improves generalization to held-out instructions. In addition, our error analysis suggests that questions in K2Q provide enhanced contextualization compared to simple ones, resulting in more grounded answers from models, regardless of correctness. In future work, we plan to create templates covering a wider range of question types, few-shot instances, chain-of-thought answers with layout-informed explanations, and multi-round instructions. We hope this work encourages researchers to carefully consider the data used for generative modeling.

Limitations

We conducted our experiments using three trainable models (Donut, Pix2Struct, and DocLLM) and four zero-shot models (UDOP, mPlugDocOwl 1.5, mPlugDocOwl 1.5 chat, and GPT4). With additional resources, training state-of-the-art OCR-free models such as the mPlugDocOwl models would provide more complete results. Mainly due to resource limitations for training vision-based LLMs, we left these experiments (along with training UDOP) for future work.

Additionally, while K2Q alleviates the burden of data collection by relying on existing KIE datasets, it still requires human intervention to manually curate high-quality, diverse templates. We will investigate the use of LLMs such as GPT4 for generating such templates for VRDU applications in future work.

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A Dataset Construction

A.1 Data Cleaning

To make questions and answers sound natural, we apply several data cleaning rules to the KIE annotations in the datasets instead of using the OCR text. We describe the data cleaning method for each dataset below.

Ad-Buy. For all entities, we replace all new line characters with spaces except for addresses where we insert a comma. Next, we check if the text is enclosed in brackets (parentheses, square, braces, etc.) and if so, we remove the brackets. In addition, we found that some entities were prefixed with “REMIT TO”, and so we deleted any occurrence of this prefix. For the `advertiser`, `agency`, `product`, `program_desc`, and `tv_address` entities, we convert all multi-word, full upper case entities into title case. We have an additional caveat for `tv_address` and do not modify words less than two characters as they are likely US states.

CORD. CORD contains too many entities to opt for an entity-type approach. However, two-thirds of all entities are numeric. For these entities, we remove all OCR tokens for which over half the characters are non-numeric. We also remove any leading or trailing punctuation symbols from the entities.

Docile. We take a similar approach to cleaning Docile entities as with CORD due to the similarities in their annotation schemes. For numeric entities, we consider only numerical and punctuation symbols. We treat numbers differently in Docile than in CORD due to our dataset-specific analyses of the original annotations. For non-numeric entities, we replace newline characters with a comma and a space. Additionally, we pre-process the text to the title case. We acknowledge that this may not always be the correct choice of data cleaning.

KLC. For KLC, we only title case entities that are `address_post_town`, `address_street_line`, and `charity_name`. The other entities are identification type entities (e.g., `charity_name`) or numbers and so data cleaning may compromise their values.

Reg. Form. We apply data cleaning to Reg. Form using the same methods described for Ad-Buy. In addition, we apply additional checks for

Dataset	Train	Dev	Test
Ad-Buy	11,362 (480)	— (—)	3,757 (161)
CORD	62,948 (800)	7,242 (100)	7,551 (100)
Docile	169,664 (5,180)	— (—)	15,893 (500)
KLC	27,180 (1,729)	7,294 (440)	10,339 (609)
Reg. Form	17,539 (1,436)	— (—)	5,888 (479)

Table 8: Number of questions and documents (in brackets) for K2Q splits.

words that should not be converted to title case. For each entity type, we find common acronyms used in the text (e.g., USA, LLC) and ensure these tokens remain in all caps.

A.2 Data Splits

We follow the data splits provided by the original datasets. For Docile, the downloadable test set does not give the annotated entities; we thus use the dev set as its test set. For Ad-Buy and Reg. Form, multiple splits were given in Wang et al. (2023c) and so we follow the split in Wang et al. (2023a). We provide the number of questions and documents for each dataset split in Table 8.

B K2Q Examples

In this section, we provide example documents and questions generated from each of the five datasets of K2Q. Figure 5, Figure 6, Figure 7, Figure 8, and Figure 9 compare questions generated in K2Q to those found in InstructDoc (Tanaka et al., 2024), UReader (Ye et al., 2023b), and SD. Note that InstructDoc and UReader do not contain most of the datasets used in this work. We thus use their templates to generate questions as described in the respective papers.

The key differences between the questions found in K2Q and those found in UReader and SD are the style and informativeness of the questions. Our templates are designed with the dataset in mind and thus sound more natural and resemble questions that one might encounter in the real world. Furthermore, our templates use other key entities within the questions to better contextualize queries as well as enable a wider selection of questions. We note that for datasets such as Docile which contain line items, UReader may generate ambiguous questions.

INVOICE

Page 1 of 1

Advertiser	POL/Michael Bloomberg/President/US/D	Invoice #	2335618-1
Product	BLOOMBERG 4 PRES	Invoice Date	01/26/20
Estimate Number	0116	Invoice Month	January 2020
		Invoice Period	12/30/19 - 01/26/20
Property	KRCW	Order #	2335618
Account Executive	Telerep Philadelphia	Alt Order #	09735556
Sales Office	Telerep/Philadelphia	Deal #	
Sales Region	National	Order Flight	01/11/20 - 01/17/20
Billing Calendar	Broadcast	Agency Code	9915458
Billing Type	Cash	Advertiser Code	MBLM
Special Handling		Product 1/2	MBLM

K2Q Questions:

Q: From when until when is the contract in flight?

A: 01/11/20 - 01/17/20

Q: Did POL/Michael Bloomberg/President/US/D order the advertisement "Bloomberg 4 Pres"? A: Yes

UReader and SD Questions:

Q: What is the value for the product? A: BLOOMBERG 4 PRES

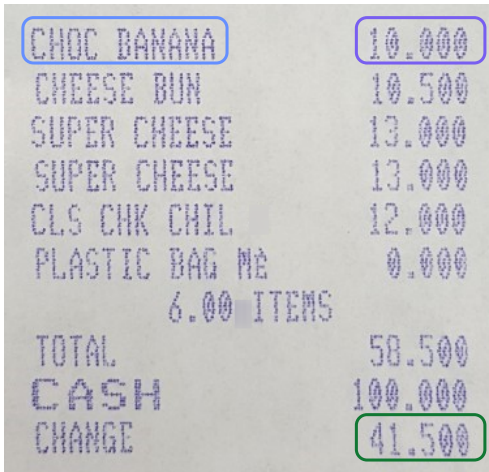
Q: What is the value for the flight start date? A: 01/11/20

InstructDoc Questions:

Q: There are 15 categories for selection: "advertiser", ..., and "tv_address". Please output the category corresponding to the text "BLOOMBERG 4 PRES".

A: product

Figure 5: Excerpt of an Ad-Buy document with generated questions from K2Q, InstructDoc, UReader, and SD.



K2Q Questions:

Q: What is the menu item that cost a total of 10.000 in this bill?
A: CHOC BANANA

Q: How much did the order(s) of CHOC BANANA cost in total?
A: 10.000

Q: Is "58.500" the amount of change in cash in this receipt? A: No

UReader and SD Questions:

Q: What is the value for the amount of change in cash? A: 41.500

Q: What is the value for the menu item? A: CHOC BANANA

InstructDoc Questions:

Q: Please tell me the category of the text "41.500" to select from following classes: "menu.nm", ..., and "total.menuqty_cnt".
A: total.changeprice

Figure 6: Excerpt of a CORD document with generated questions from K2Q, InstructDoc, UReader, and SD. Note that the second question for UReader and SD is not present in SD due to its ambiguity.

Order Details

#	Market Station	Bind To	Start Date	End Date	No Of W.	On Air W.	Sch Days	Skip W.	M	T	W	Th	F	Sa	Su	Spots/ Week	Spot Len.	Revenue Type	Rate	Ord. Spots	Ord. Cost	Make Good
1	Fort Collins KCOL-AM	06:00-10:00 Commercial	06/13/2022	06/16/2022	1	1	3	0	X	X	X	X	-	-	-	20	60	Local Agency-Political	\$44.00	20	\$880.00	
2	Fort Collins KCOL-AM	10:00-15:00 Commercial	06/13/2022	06/16/2022	1	1	3	0	X	X	X	X	-	-	-	20	60	Local Agency-Political	\$39.00	20	\$780.00	
3	Fort Collins KCOL-AM	15:00-19:00 Commercial	06/13/2022	06/16/2022	1	1	3	0	X	X	X	X	-	-	-	20	60	Local Agency-Political	\$31.00	20	\$620.00	

# Digital	Start Date	End Date	Description	Rev. Type	Impressions
1	06/13/2022	06/16/2022	iHeart Audience Network (iAN) - Streaming	POLITICAL	31250

Number of Spots:	60	Ordered Gross:	\$2,280.00
Number of Miscellaneous Lines:	0	Agency Commission:	\$342.00
Number of Digital Impressions:	31250	Ordered Net:	\$1,938.00
		Digital Assets Gross:	\$500.00
		Agency Commission:	\$75.00
		Digital Assets Net:	\$425.00
		Total Net Due:	\$2,363.00

K2Q Questions:

Q: How much is the **total amount with tax** of the 2nd item? A: 780.00
 Q: What is the **total amount to be paid** in the document? A: 2,363.00
 Q: Is **"60.48"** the total amount to be paid in this document? A: No

UReader and SD Questions:

Q: What is the value for the **total amount with tax**? A: 780.00
 Q: What is the value for the **total amount to be paid**? A: 2,363.00

InstructDoc Questions:

Q: The document contains 36 key categories: "account_num", ..., and "total". Kindly identify the category related to the text **"2,363.00"** mentioned in the provided document.
 A: amount_due

Figure 7: Excerpt of a Docile document with generated questions from K2Q, InstructDoc, UReader, and SD. Note that the first question for UReader and SD is not present in SD due to its ambiguity.

HASLEMERE EDUCATIONAL MUSEUM

REPORT OF THE TRUSTEES

FOR THE YEAR ENDED 31ST DECEMBER 2014

The Trustees who are also directors of the charity for the purposes of the Companies Act 2006 report with the financial statements of the charity for the year ended 31st December 2014 adopted the provisions of the Statement of Recommended Practice (SORP) 'Accredited Charities' issued in March 2005.

REFERENCE AND ADMINISTRATIVE DETAILS

Registered Company number:
03203702 (England and Wales)

Registered Charity number:
1071244

Registered Office: 78 High Street
Haslemere
Surrey
GU27 2LA

K2Q Questions:

Q: What is the number of the charity Haslemere Educational Museum? A: 1071244

Q: What is the postcode for Haslemere Educational Museum? A: GU27 2LA

Q: Is GU27 2LA the postcode of street 78 High Street? A: Yes

InstructDoc Questions:

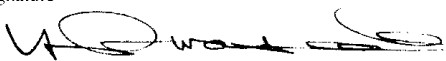
Q: Categories: "address_postcode", ..., and "spending_annually_in_british_pounds". Kindly provide me with the category of the text "78 High Street". A: address_street_line

UReader and SD Questions:

Q: What is the value for the charity number? A: 1071244

Q: What is the value for the address postcode? A: GU27 2LA

Figure 8: Excerpt of a KLC document with generated questions from K2Q, InstructDoc, UReader, and SD.

14. Have two copies of this material been filed with the Department of Justice? Yes ☒ No ☐		
15. Has this material been labeled as required by the act? Yes ☒ No ☐		
Date of report 8/21/91	Name and title Yasumasa Iwasaki Executive Director	Signature 

K2Q Questions:

Q: What position does the form signer hold? A: Executive Director

Q: Did Yasumasa Iwasaki sign the form? A: Yes

InstructDoc Questions:

Q: Options: "file_date", ..., and "signer_title". Please select the category associated with the text "Yasumasa Iwasaki" in the given document.

A: amount_due

UReader and SD Questions:

Q: What is the value for the signer title? A: Executive Director

Q: What is the value for the signer name? A: Yasumasa Iwasaki

Figure 9: Excerpt of a Reg. Form document with generated questions from K2Q, InstructDoc, UReader, and SD.

For example, the question regarding the line item total price in Figure 7 has a variety of answers that could be correct as it is unclear which line was being referred to. Docile contains documents that have many lines, so such under-specified questions are avoided.

The primary difference between K2Q and InstructDoc lies in the kind of questions being asked. As explained in Section 3.2.1, InstructDoc transforms the information extraction problem into a classification problem. Therefore, the KIE portion of InstructDoc trains and tests a model’s ability to characterize a key entity to its correct type, but not to find the entity itself. Other than formulating questions for a different task, InstructDoc has five dataset-agnostic templates it populates. Consequently, the variety of questions in K2Q is much wider.

C Experimental Set-up

C.1 Model Training Details

Each trainable model (Donut, Pix2Struct, and DocLLM) is then fine-tuned for ten epochs on each dataset of this study individually, with a learning rate of 10^{-4} . Donut and Pix2Struct use the AdaFactor optimizer with a cosine scheduler with 500 warm-up steps and batch sizes of four and one respectively. A weight decay of 10^{-5} is used. DocLLM uses the AdamW optimizer with a cosine scheduler with warm-up, and a batch size of 24 through gradient accumulation. We use available implementations of these models (Wolf et al., 2019) in PyTorch 1.13 (Paszke et al., 2017). Each model is fine-tuned on a single A10G GPU.

C.2 Metric Definitions

In this section, we provide more details regarding the perplexity, Self-BLEU, and ANLS scores used in Section 3.2.1 and Section 4 respectively.

Perplexity. Perplexity is a measure of how likely a language model is to generate a given sequence of text. Mathematically, it is the inverse of the joint probability of a sequence of text being drawn from a distribution. Alternatively, it is the exponentiated average of the negative log-likelihood of the input probability conditioned on all previous tokens. If we consider a tokenized string $s = \{s_1, \dots, s_n\}$, the perplexity of a model M is defined as

$$\text{PPL}(s) \stackrel{\text{def}}{=} \exp\left(-\frac{1}{n} \sum_{i=1}^n \log p_M(s_i | s_{<i})\right)$$

where $p_M(s_i | s_{<i})$ is the probability of witnessing token s_i in the model, after observing all past tokens of s .

Self-BLEU. Self-BLEU (Zhu et al., 2018) measures the lexical diversity of a corpus of text. It leverages the BLEU metric (Papineni et al., 2002) to evaluate how one sentence resembles the rest in a collection. It is obtained by averaging the BLEU score of each sentence of the corpus (hypothesis) with the rest of the collection (reference). A sample of the corpus is typically used as the computation is too resource-intensive otherwise; we use a sample size of 5,000 test questions per dataset in this work.

ANLS. The Average Normalised Levenshtein Similarity is a string edit distance that measures the number of edits it takes to transform one string into another, normalized against string length. It is defined as

$$\text{ANLS}(s, t) \stackrel{\text{def}}{=} \begin{cases} 1 - \text{NL}(s, t) & \text{if } \text{NL}(s, t) < \tau \\ 0 & \text{otherwise} \end{cases}$$

where $\text{NL}(s, t)$ is the normalized Levenshtein Distance between s and t , and τ is a distance threshold (typically 0.5). The reported score over a dataset is the mean ANLS score over all answers in a dataset.

D Additional Experimental Results

In this section, we provide the results of additional experiments that we could not provide in the main paper due to space constraints.

D.1 Results on SD

Table 9 gives the ANLS scores of our different modeling set-ups on SD. As one would expect, training on SD yields the best outcomes. However, as observed in Section 4.4, DocLLM performs impressively when fine-tuned using the more diverse templates of K2Q. This may suggest that the jump in model size from Donut and Pix2Struct to DocLLM enables better generalization with high-quality data. We note that like Table 7, Pix2Struct trained on Docile is the only time we observe an improvement in training with SD rather than K2Q.

D.2 Ablation Studies

K2Q contains several interesting properties such as the existence of extractive and boolean questions and questions that contain multiple entities. We

Model	Ad-Buy	CORD	Docile	KLC	Reg-Form
Dnt _{ZS}	4.8	61.1	8.9	7.6	15.5
P2S _{ZS}	19.1	63.8	25.2	12.8	30.0
Doc _{ZS}	1.6	72.9	28.1	78.2	2.8
UDOP _{ZS}	4.9	44.2	21.6	12.8	4.8
mPD _{ZS}	64.4	55.2	41.8	71.6	60.5
mPDC _{ZS}	65.9	42.5	41.2	71.8	59.8
GPT4 _{ZS}	50.5	82.4	48.0	38.7	70.2
Dnt _{SD}	61.2	90.2	39.9	55.9	78.3
P2S _{SD}	79.0	89.3	61.2	76.3	85.6
Doc _{SD}	95.7	94.4	86.7	88.9	66.7
Dnt _{mK2Q}	24.0	80.3	19.2	24.1	38.8
P2S _{mK2Q}	27.7	81.7	27.6	29.0	33.9
Doc _{mK2Q}	91.1	94.3	82.1	85.6	62.6
Dnt _{K2Q}	24.2	81.7	19.3	24.3	35.9
P2S _{K2Q}	28.2	81.6	27.6	29.9	28.2
Doc _{K2Q}	95.2	95.9	82.5	87.2	66.9

Table 9: ANLS ($\uparrow$) results on SD test set using various training settings for models Donut (Dnt), Pix2Struct (P2S), DocLLM (Doc), UDOP, mPlugDocOwl 1.5 (mPD), mPlugDocOwl 1.5-Chat (mPDC), and GPT4.

conduct an ablation study on how different training configurations and different models perform different question types and number of entities. Number of entities is counted as number of entities used within the question. Therefore, the minimum number of entities for extractive questions is zero while the minimum number of entities for boolean questions is one. These are given in Table 10, Table 11, Table 12, Table 13, and Table 14. Observe that Pix2Struct and Donut are unable to answer any boolean questions under the zero-shot setting or when trained on SD. This demonstrates the difficulty of generalizing beyond the types of questions seen during training. In general, the performance of the model decreases as the number of entities in the question increases.

D.3 Extended Quantitative Comparison of Templates

We give more results for the experiments described in Section 4.4. Figure 10 gives a complementary figure to Figure 3 but using K2Q rather than mK2Q with similar patterns observed. Since both Figure 3 and Figure 10 display relative differences, we illustrate the ANLS scores used to derive them in Figure 11 and Figure 12.

D.4 Extended Evaluation of Generated Errors

We give a breakdown of the groundedness and types of errors exhibited by our fine-tuned models on Ad-Buy, CORD, Docile, and Reg. Form

in Figure 13, Figure 14, Figure 15, and Figure 16 respectively. We note that using K2Q does not always yield better groundedness than SD. This is likely due to the clarity of simple templates asking for an entity, specifically for CORD and Docile, for which there are many different entities. Nevertheless, we observe that for the majority of datasets, training using a version of K2Q leads to higher groundedness as well as a higher number of misprints. Misprints are a good error to observe as it indicates that the model was able to understand the question and answer, but could not generate the exact text.

We provide a few concrete examples of errors observed in Reg. Form below.

D.4.1 Error Examples.

We analyze some of the errors made by DocLLM (the best-performing model) on Reg. Form as a guide for future model development. Given that K2Q closely resembles the kind of questions users in the wild are likely to ask, such an error analysis provides insights as to what errors downstream applications can expect. An analysis using SD would not be able to provide such insights.

A representative example of an incorrect answer is from a question “Who is the signer of the form?” in which the OCR output for one of the documents is “/ Chad Horrell” while the generated answer is “Chad Horrell”. We noticed incorrect answers are often subsequences of the ground truth answer or token sequences with a slight offset from the ground truth. In the above example, the solution may be to start with higher quality annotations as the “/” seems out of place. Another example of what seems to be a misprint error is the question “What is the registration ID of this form?” which produces the answer “6228” instead of the ground truth “6278”.

We also observe many errors that seem to be unrelated to the question or document. For instance, DocLLM generated the answer “Government Of Japan–Japan External Trade Organization” instead of “Embassy Of The State Of Qatar” for the question, “Which foreign principal is this form about?”. The generated answer is not present in the document but is present in a different document of the training dataset. This could suggest that the models may suffer from overfitting or hallucination issues.

We note also that numbers and dates can be reformatted in the generated output which can lead to ungrounded outputs with low ANLS. We eval-

Model	All	Extractive Questions					Boolean Questions		
		All	0 Entities	1 Entity	2 Entities	3 Entities	All	1 Entity	2 Entities
Donut _{ZS}	1.4	2.5	2.9	1.7	0.6	19.0	0.0	0.0	0.0
Pix2Struct _{ZS}	13.5	24.1	26.2	15.9	26.4	32.6	0.0	0.0	0.0
DocLLM _{ZS}	23.9	5.3	5.6	2.2	7.4	16.8	47.6	47.3	48.1
UDOP _{ZS}	29.1	22.0	16.4	23.4	36.1	42.4	38.2	42.8	27.7
mPlugDO _{ZS}	60.9	47.2	52.2	31.0	54.0	25.8	78.4	82.2	69.6
mPlugDOC _{ZS}	61.9	53.1	57.0	38.9	59.2	44.2	73.2	76.4	65.9
GPT4 _{ZS}	72.7	68.1	65.2	65.6	79.6	76.7	78.6	76.0	84.5
Donut _{SD}	7.7	13.7	18.9	6.9	5.7	7.5	0.0	0.0	0.0
Pix2Struct _{SD}	18.0	32.2	37.4	19.7	32.9	13.3	0.0	0.0	0.0
DocLLM _{SD}	46.3	52.2	60.2	35.7	45.8	61.5	38.7	38.9	38.2
Donut _{mK2Q}	56.4	48.2	44.6	44.6	62.8	65.1	66.9	69.7	60.5
Pix2Struct _{mK2Q}	69.2	64.5	64.3	61.7	68.6	65.8	75.2	79.0	66.5
DocLLM _{mK2Q}	92.6	89.3	93.6	83.1	84.2	77.5	96.7	97.6	94.8
Donut _{K2Q}	58.5	49.1	45.9	44.8	63.4	63.1	70.5	73.9	62.6
Pix2Struct _{K2Q}	73.8	68.6	69.6	63.2	72.1	69.2	80.3	84.7	70.2
DocLLM _{K2Q}	93.9	91.5	96.2	85.6	85.3	77.4	96.9	97.6	95.2

Table 10: Ablation study of ANLS ($\uparrow$) for K2Q Ad-Buy.

Model	All	Extractive Questions			Boolean Questions
		All	0 Entities	1 Entity	
Donut _{ZS}	18.0	36.3	62.1	26.9	0.0
Pix2Struct _{ZS}	27.0	54.3	68.7	49.1	0.0
DocLLM _{ZS}	43.0	62.9	83.7	55.4	23.4
UDOP _{ZS}	29.7	33.0	49.0	27.2	26.4
mPlugDO _{ZS}	64.5	76.8	72.9	78.2	52.3
mPlugDOC _{ZS}	64.4	76.2	74.0	77.0	52.8
GPT4 _{ZS}	85.3	87.7	88.0	87.6	82.9
Donut _{SD}	24.4	49.0	86.1	35.5	0.0
Pix2Struct _{SD}	31.0	62.4	89.3	52.6	0.0
DocLLM _{SD}	43.6	70.8	95.6	61.9	16.6
Donut _{mK2Q}	82.7	83.8	84.2	83.6	81.7
Pix2Struct _{mK2Q}	84.7	88.4	86.6	89.0	81.1
DocLLM _{mK2Q}	94.0	95.1	96.8	94.5	93.0
Donut _{K2Q}	83.3	85.1	84.3	85.4	81.5
Pix2Struct _{K2Q}	86.5	90.2	88.8	90.6	83.0
DocLLM _{K2Q}	96.5	96.6	96.9	96.5	96.3

Table 11: Ablation study of ANLS ($\uparrow$) for K2Q CORD.

uated this case and found that this type of error occurs in less than 0.5% of cases across datasets and models in the fine-tuned setting. Therefore, we do not categorize such errors here and place them in **other**.

Model	All	Extractive Questions			Boolean Questions
		All	0 Entities	1 Entity	
Donut _{ZS}	7.2	10.4	11.2	0.2	0.0
Pix2Struct _{ZS}	24.1	35.0	37.0	10.1	0.0
DocLLM _{ZS}	48.0	42.1	43.5	23.9	61.0
UDOP _{ZS}	35.1	33.2	34.7	14.0	39.2
mPlugDO _{ZS}	56.0	49.7	52.2	18.0	69.7
mPlugDOC _{ZS}	53.4	50.4	53.0	17.5	59.9
GPT4 _{ZS}	61.2	54.0	54.5	47.0	77.3
Donut _{SD}	22.3	32.4	34.5	5.2	0.0
Pix2Struct _{SD}	35.5	51.6	53.0	33.1	0.0
DocLLM _{SD}	53.0	76.4	76.9	70.0	1.5
Donut _{mK2Q}	47.7	39.0	40.9	14.4	66.9
Pix2Struct _{mK2Q}	59.0	60.1	60.7	51.8	56.7
DocLLM _{mK2Q}	88.9	85.7	85.8	84.5	95.8
Donut _{K2Q}	47.8	39.1	41.0	14.4	67.2
Pix2Struct _{K2Q}	59.5	60.6	61.2	53.4	57.1
DocLLM _{K2Q}	90.0	87.0	87.2	83.7	96.8

Table 12: Ablation study of ANLS ($\uparrow$) for K2Q Docile.

Model	All	Extractive Questions			Boolean Questions		
		All	0 Entities	1 Entity	All	1 Entity	2 Entities
Donut _{ZS}	3.8	9.2	19.9	5.1	0.1	0.3	0.1
Pix2Struct _{ZS}	11.8	28.8	38.7	25.0	0.0	0.0	0.0
DocLLM _{ZS}	80.4	80.4	85.1	78.7	80.4	80.1	80.5
UDOP _{ZS}	30.0	22.2	31.5	18.7	35.4	50.7	30.6
mPlugDO _{ZS}	66.8	65.6	66.7	65.2	67.6	78.4	64.2
mPlugDOC _{ZS}	66.0	64.9	64.1	65.3	66.7	81.0	62.4
GPT4 _{ZS}	68.1	65.7	53.4	70.3	69.8	62.1	72.2
Donut _{SD}	15.6	37.9	66.7	27.0	0.0	0.1	0.0
Pix2Struct _{SD}	28.0	68.4	89.2	60.5	0.0	0.0	0.0
DocLLM _{SD}	62.8	88.7	95.8	86.0	44.8	47.2	44.0
Donut _{mK2Q}	67.2	45.9	75.6	34.6	82.1	86.8	80.6
Pix2Struct _{mK2Q}	77.0	72.4	90.7	65.5	80.2	81.3	79.9
DocLLM _{mK2Q}	92.6	88.4	94.3	86.2	95.5	98.2	94.7
Donut _{K2Q}	68.1	46.1	76.0	34.8	83.3	87.8	82.0
Pix2Struct _{K2Q}	79.4	74.0	91.5	67.3	83.2	84.1	82.9
DocLLM _{K2Q}	93.6	89.7	95.8	87.4	96.3	98.9	95.4

Table 13: Ablation study of ANLS ($\uparrow$) for K2Q KLC.

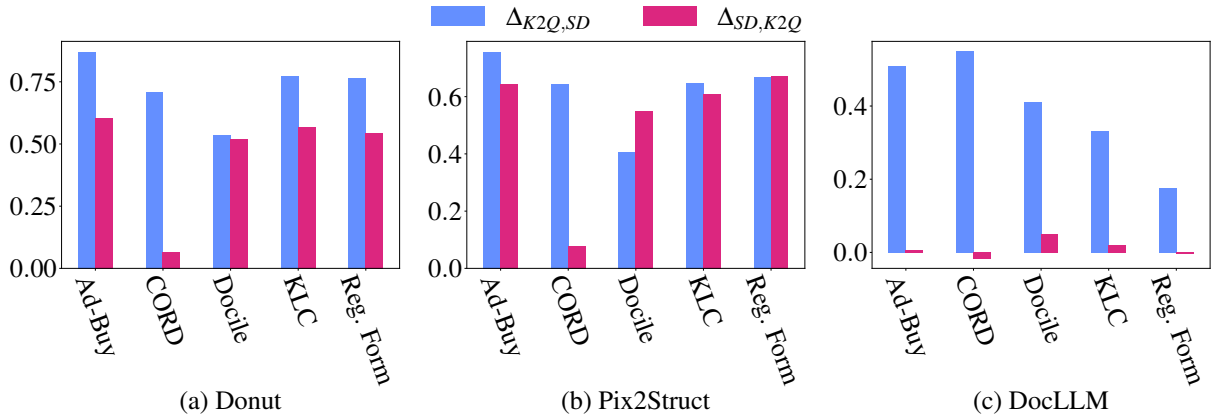


Figure 10: Comparison of training and evaluating on complex questions (K2Q) and simple questions (SD).

Model	All	Extractive Questions			Boolean Questions		
		All	0 Entities	1 Entity	All	1 Entity	2 Entities
Donut _{ZS}	5.9	15.8	15.2	25.0	0.0	0.0	0.0
Pix2Struct _{ZS}	24.0	63.7	62.1	91.3	0.0	0.0	0.0
DocLLM _{ZS}	27.6	4.2	4.1	5.0	41.7	41.8	40.3
UDOP _{ZS}	39.1	19.0	16.9	53.5	51.2	51.6	45.3
mPlugDO _{ZS}	65.2	62.0	60.1	93.0	67.1	66.3	78.8
mPlugDOC _{ZS}	66.8	65.5	63.9	92.2	67.6	67.4	71.2
GPT4 _{ZS}	76.5	74.6	73.2	97.1	77.7	78.4	66.1
Donut _{SD}	19.7	52.4	51.0	75.3	0.0	0.0	0.0
Pix2Struct _{SD}	29.7	78.8	78.3	87.0	0.0	0.0	0.0
DocLLM _{SD}	66.2	64.2	62.6	91.1	67.5	68.1	58.5
Donut _{mK2Q}	81.4	69.7	69.6	71.5	88.5	89.3	77.1
Pix2Struct _{mK2Q}	87.0	81.1	80.9	84.3	90.7	91.1	84.7
DocLLM _{mK2Q}	78.0	62.3	60.7	88.5	87.4	87.1	91.9
Donut _{K2Q}	82.7	70.7	70.8	69.3	90.0	90.4	83.1
Pix2Struct _{K2Q}	88.7	82.7	82.7	82.5	92.3	92.5	89.4
DocLLM _{K2Q}	80.3	66.2	64.7	91.0	88.9	88.4	95.8

Table 14: Ablation study of ANLS ($\uparrow$) for K2Q Reg. Form.

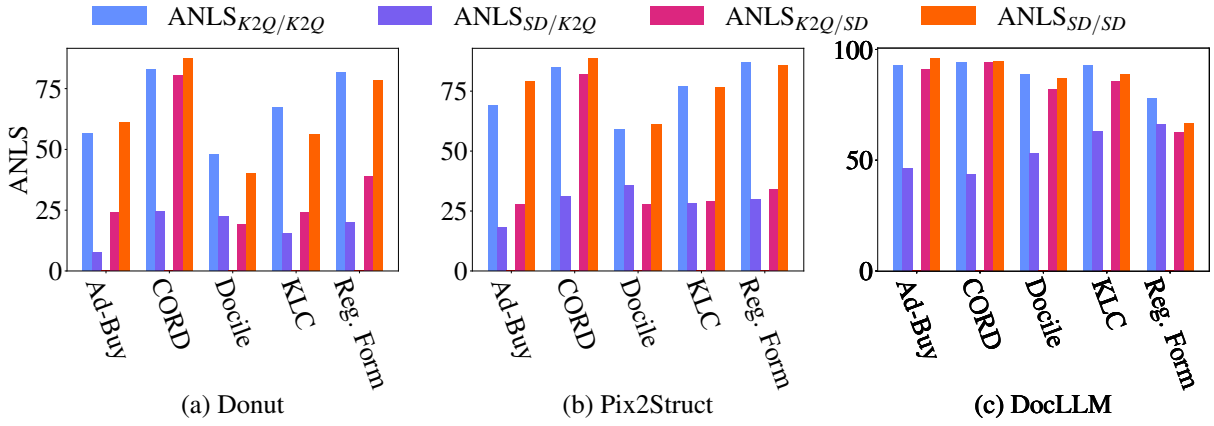


Figure 11: Comparison of ANLS scores for training and evaluating on sampled complex questions (mK2Q) and simple questions (SD).

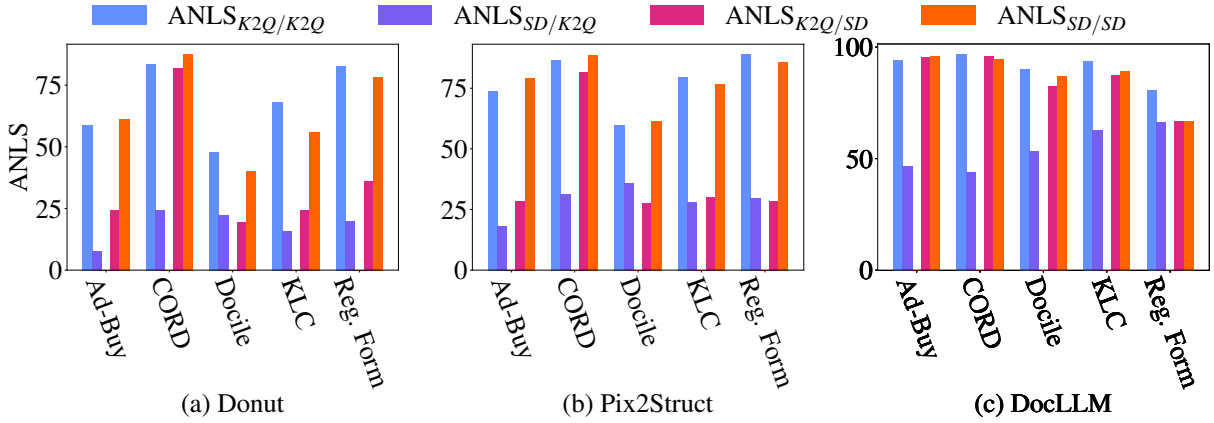


Figure 12: Comparison of ANLS scores for training and evaluating on complex questions (K2Q) and simple questions (SD).

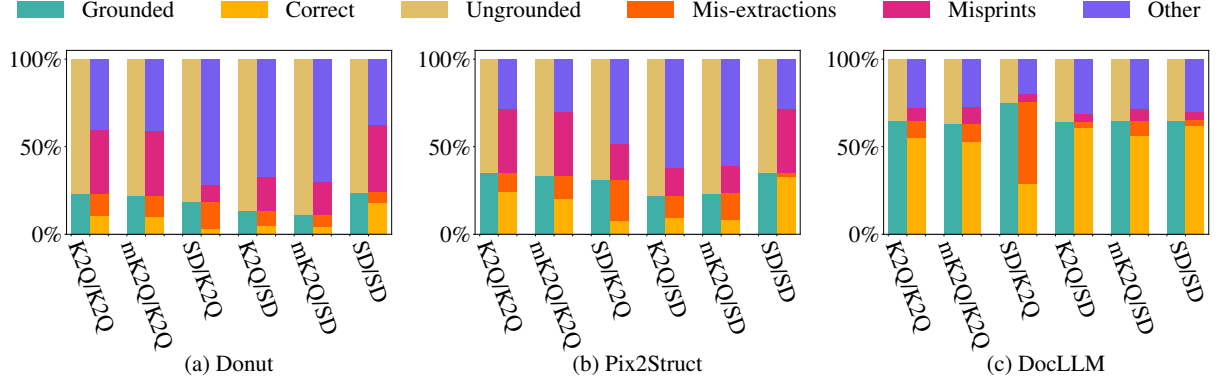


Figure 13: Detailed breakdown of groundedness and error types for Ad-Buy using different training / testing datasets.

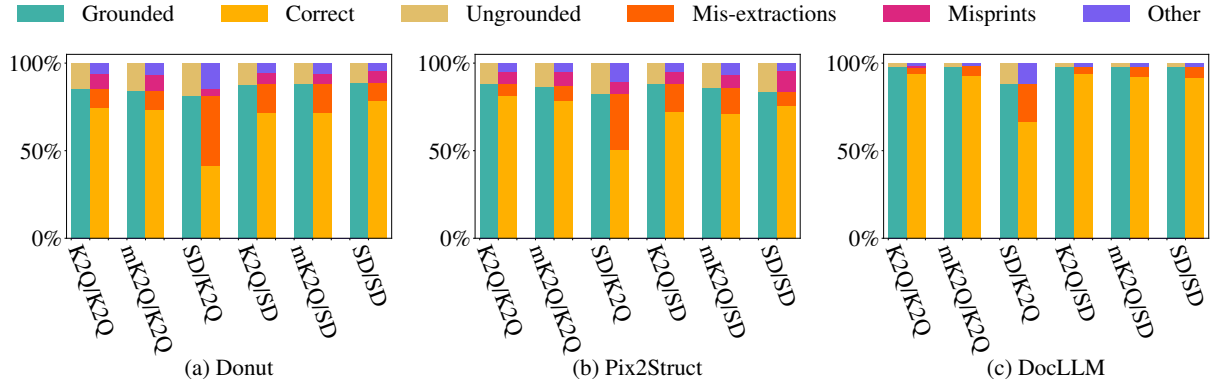


Figure 14: Detailed breakdown of groundedness and error types for CORD using different training / testing datasets.

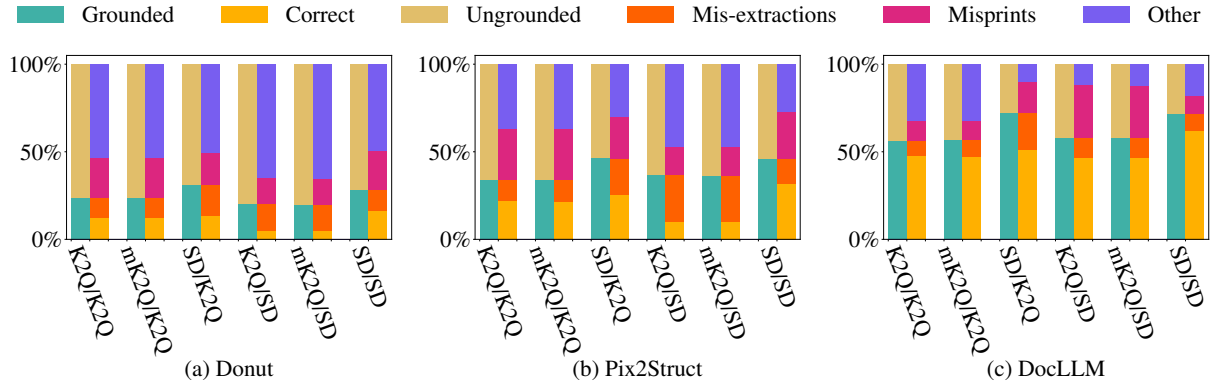


Figure 15: Detailed breakdown of groundedness and error types for Docile using different training / testing datasets.

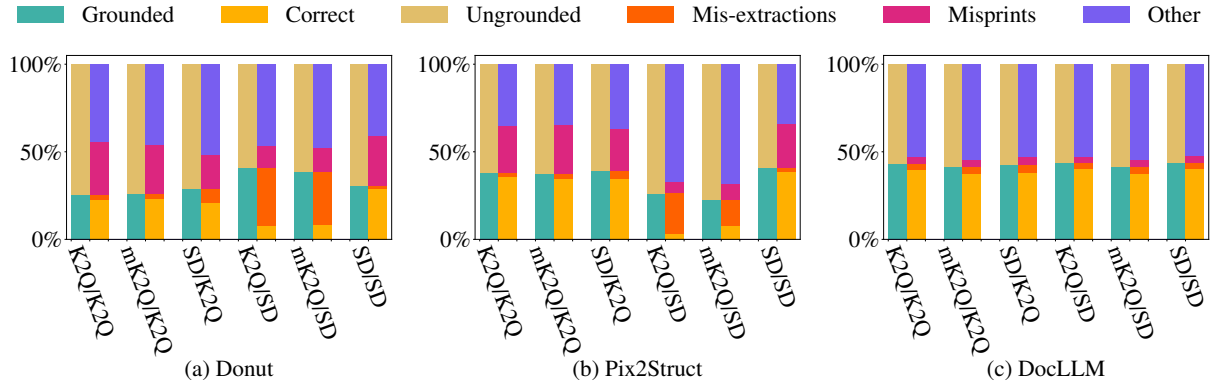


Figure 16: Detailed breakdown of groundedness and error types for Reg. Form using different training / testing datasets.